
TerraBit: Low-Bit Quantization of Geospatial Embedding Lakes

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Abstract

Geospatial foundation models produce planet-scale *embedding lakes*—large, centralized stores of precomputed vectors analogous to data lakes—whose storage and retrieval costs dominate deployment budgets. Recent systems distribute global embedding fields at `int8` precision, but the quality–cost trade-off below 8 bits remains unexplored for remote sensing embeddings. We present **TerraBit**, a post-hoc quantization study spanning `float16` through 1-bit binary on 49,785,116 real 1024D embeddings (*183 GiB*), evaluated with label-free neighborhood-preservation metrics grounded in intrinsic-dimension analysis. `Int4` retains 91% kNN recall at $6.3\times$ compression; binary achieves $16.5\times$ compression with 65% recall, sufficient for coarse-stage candidate generation. We additionally benchmark TurboQuant [Zandieh et al., 2025], finding that its random orthogonal pre-rotation helps most in the ultra-low-bit regime: 2-bit recall rises from 0.54 to 0.71, while the advantage shrinks to near zero by 8 bits.

1 Introduction

Geospatial foundation models [Mendieta et al., 2023, Jakubik et al., 2024, Clay Foundation, 2025] produce high-dimensional embeddings for retrieval, change detection, deduplication, and spatial indexing, forming *embedding lakes*—large, centralized vector stores analogous to data lakes. At planet scale, storage and memory bandwidth—not model accuracy—become the deployment bottleneck. A corpus of 50M embeddings at $d=1024$ in `float32` occupies *183 GiB*; halving per-vector cost saves nearly *100 GiB* with no retraining.

Several recent systems highlight this tension. AlphaEarth releases global embedding fields cast to `int8` via post-hoc calibration [Brown et al., 2025]. TESSERA applies quantization-aware training to a learned 128D bottleneck and distributes the result at `int8` [Feng et al., 2025]. ESD [Chen et al., 2026] takes a learned-compression approach, training a VQ-VAE-style autoencoder with FSQ bottlenecks [Mentzer et al., 2024] to compress Landsat+MODIS time series into `uint16` latent vectors, achieving $\sim 340\times$ compression while improving downstream land-cover classification over raw reflectance (79.7% vs. 76.9%). These systems treat `int8`—or learned discrete codes—as the practical floor. **TerraBit** asks: *how far below `int8` can we push post-hoc quantization while preserving nearest-neighbor structure?*

We sweep seven bit-widths from `float16` to 1-bit binary on a real 49.8M-embedding corpus and measure neighborhood preservation with label-free metrics. Figure 2 shows graceful degradation in local-neighborhood preservation, and Table 1 quantifies recall@10 trade-offs for cosine and Euclidean distance. The frontier is smooth: `int4` retains 91% kNN recall at $6.3\times$ compression, while binary reaches $16.5\times$ with recall sufficient for shortlist retrieval [Wang et al., 2018].

Our contributions are: (1) an intrinsic-dimension-guided evaluation framework combining ID diagnostics [Levina and Bickel, 2004, Facco et al., 2017] with label-free retrieval fidelity metrics;

(2) a systematic post-hoc quantization study (float16, fp8, int8, int4, int3, int2, binary, and TurboQuant at four bit-widths) on 49.8M real geospatial embeddings; (3) a schema-preserving parallel Parquet transformation pipeline that processes the full corpus in 45 min at 18,359 rows/s; (4) a retrieval benchmark showing $>1,300$ QPS on GPU with recall preserved across hardware; and (5) an empirical comparison showing that rotation-based quantization (TurboQuant) improves recall most at 2–3 bits, with diminishing gains as precision increases.

2 Related Work

Geospatial foundation models. Recent work trains large vision encoders on satellite imagery and releases pretrained embeddings as data products [Mendieta et al., 2023, Jakubik et al., 2024, Clay Foundation, 2025]. AlphaEarth [Brown et al., 2025] and TESSERA [Feng et al., 2025] both distribute global int8 embedding fields, recognizing that storage cost matters as much as model quality at deployment scale.

Vector quantization and retrieval. Product quantization [Jégou et al., 2011], learned anisotropic quantization [Guo et al., 2020], and locality-sensitive hashing [Indyk and Motwani, 1998] compress embeddings for approximate nearest-neighbor search. FAISS [Johnson et al., 2021] implements these at billion scale. LLM.int8() [Dettmers et al., 2022] demonstrates that aggressive quantization preserves model quality in language; Matryoshka representations [Kusupati et al., 2022] show that embedding dimensions can be truncated with graceful degradation. TurboQuant [Zandieh et al., 2025] applies a random orthogonal pre-rotation before symmetric quantization to approach rate-distortion bounds in online vector quantization. For binary coding, prior work spans angular quantization for cosine retrieval [Gong et al., 2012], deep visual-semantic quantization with learned codebooks [Cao et al., 2017], and autoencoder-based near-lossless binarization in NLP [Tissier et al., 2019]. Orthogonally, learned codebook methods such as FSQ [Mentzer et al., 2024] replace VQ-VAE codebook lookup with per-dimension scalar rounding, enabling end-to-end discrete compression without codebook collapse—an approach recently applied to geospatial embeddings by ESD [Chen et al., 2026]. At the storage layer, Parquet’s BYTE_STREAM_SPLIT encoding reorganizes floating-point byte planes before compression, while specialized numerical codecs such as Pcodec [Loncaric et al., 2025] target entropy structure beyond generic LZ codecs.

Positioning. Prior vector-compression work targets web-scale text/image embeddings or model-weight quantization; learned approaches like ESD [Chen et al., 2026] train task-specific autoencoders. TerraBit focuses on the complementary, less-studied regime of *post-hoc* quantization of geospatial embedding lakes, where spatial metadata must be preserved alongside vectors, no retraining is required, and label-free evaluation is the practical reality.

3 Method

Problem setup. Given a corpus $X = \{x_i\}_{i=1}^N$, $x_i \in \mathbb{R}^d$, stored in partitioned Parquet files with spatial/temporal metadata columns, we apply a quantizer $\hat{x}_i = Q_m(x_i)$ under method m and evaluate each method by storage footprint, neighborhood preservation, and end-to-end throughput. Our corpus comprises 49,785,116 embeddings ($d=1024$) produced by the Clay v1.5 foundation model [Clay Foundation, 2025], a ViT trained via masked autoencoding on multi-sensor satellite imagery.

Intrinsic-dimension diagnostics. Before compression, we estimate the intrinsic dimension (ID) of the embedding manifold using MLE [Levina and Bickel, 2004], TwoNN [Facco et al., 2017], and local PCA (LPCA) participation ratio across preprocessing variants (raw, centered, L2-normalized). ID estimates guide expectations for viable compression depth: if the data manifold has effective dimension $\hat{d} \ll d$, aggressive quantization should not collapse local neighborhood structure.

Quantization methods. We evaluate seven post-hoc methods spanning 16 bits to 1 bit: float16 (IEEE half-precision truncation); fp8 E4M3 (8-bit floating point with 4-bit exponent); int b for $b \in \{8, 4, 3, 2\}$ (per-dimension affine quantization with stored scale/zero-point vectors, followed by fixed-width bit packing for $b < 8$); and binary (sign-bit encoding, $\text{sign}(x_i) \in \{0, 1\}^d$, packed to $d/8$ bytes, supporting Hamming-distance search [Wang et al., 2018]).

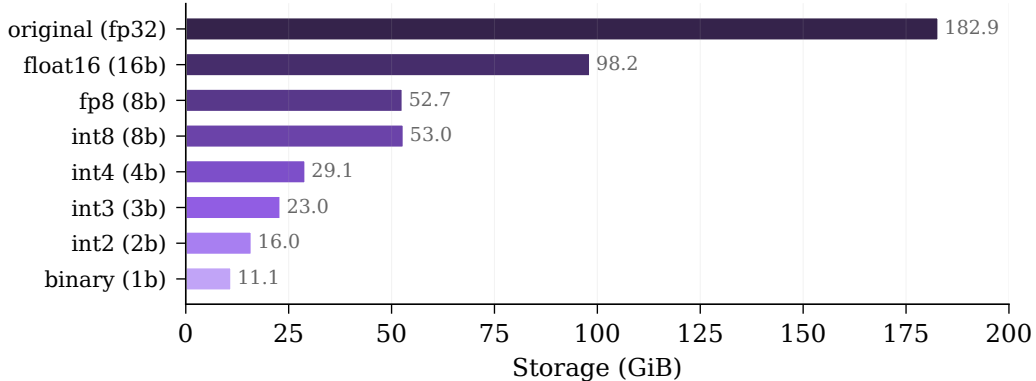


Figure 1: **Storage footprint per quantization method.** Original float32 corpus occupies *183 GiB*. Binary reduces this to *11 GiB* (16.5 $\times$); int4 to *29 GiB* (6.3 $\times$).

We additionally evaluate **TurboQuant** [Zandieh et al., 2025] at $b \in \{2, 3, 4, 8\}$ bits (turbob). TurboQuant applies a fixed random orthogonal rotation $R \in \mathbb{R}^{d \times d}$ (sampled once from a Haar-distributed ensemble via QR decomposition) before symmetric affine quantization: $\hat{x}_i = R^\top Q_b(Rx_i)$. The rotation is intended to reduce inter-channel variance disparity, improving per-dimension range utilization.

Evaluation metrics. We report label-free, retrieval-oriented metrics: **kNN recall@k** ($k \in \{10, 25, 50\}$) under cosine and Euclidean distance, measuring the fraction of true k -nearest neighbors preserved after quantization; **rank correlation of cosine similarities**, the Spearman rank correlation of pairwise cosine similarities (original vs. quantized); and **reconstruction cosine similarity** and MSE (when dequantization is available).

Transformation pipeline. The pipeline processes Hive-partitioned Parquet files in parallel. For each file, it reads the embedding column, applies quantizers, replaces only that column, and writes output preserving directory structure, non-embedding columns, and schema. Quantization parameters (scale, zero-point, method, bit-width) are stored in Parquet footer metadata for downstream decompression.

Storage-layer encoding sweep. Quantization reduces numeric precision, but file format choices can further reduce footprint without changing values. We therefore evaluate a second, lossless storage axis on a shared 100,000-embedding reservoir sample. For Parquet-native storage, we compare Snappy, Zstd, and Zstd with BYTE_STREAM_SPLIT (with dictionary encoding disabled, following Arrow’s Parquet writer semantics). For external codecs, we compare raw buffers, raw+Zstd, and Pcodec [Loncaric et al., 2025] under both row-major and dimension-major layouts. We report effective bits per original dimension, write/read throughput, and dequantization overhead, keeping quantized values fixed across storage backends.

4 Experiments

Dataset. We evaluate on 49,785,116 embeddings ($d=1024$, float32) from a geospatial foundation model, stored across 1,021 Parquet files totaling *183 GiB*. Files are Hive-partitioned by level-2 geohash (511 cells), year, and month. Each record carries spatial metadata (geohash, bounding box, WKB geometry), temporal metadata, and provenance identifiers alongside the embedding vector.

Protocol. We evaluate along two axes: quality and systems efficiency. For quality, we compute label-free fidelity metrics on a 20,000-row reservoir sample, i.e., a uniform random sample maintained with reservoir sampling (reconstruction metrics are exact over all 49,785,116 rows). For systems, we run the full-corpus transformation over all 7 methods with 8 parallel workers, reporting wall-clock time and I/O throughput.

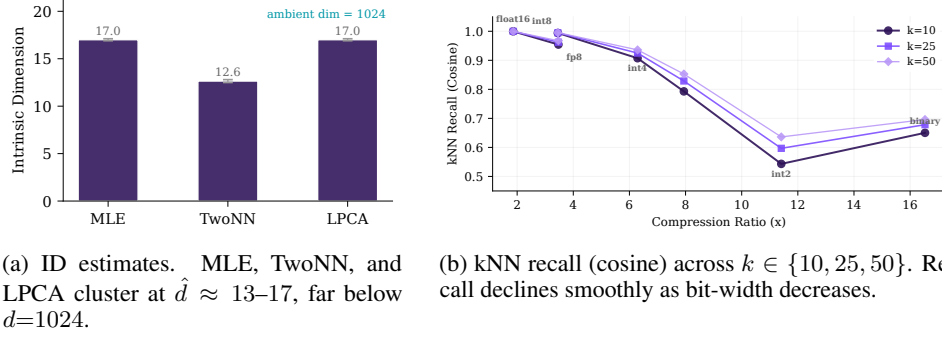


Figure 2: **Intrinsic dimension and neighborhood preservation.** Low intrinsic dimension motivates aggressive quantization, and recall degrades gracefully across operating points.

Table 1: **Quality–compression results on 49,785,116 embeddings.** Reconstruction cosine is exact over all rows; kNN recall and cosine correlation are sampled from a 20,000-row reservoir. The best compression ratio is shown in **bold**.

Method	Ratio ($\times$)	Recon. Cos $\uparrow$	C@10 $\uparrow$	E@10 $\uparrow$	Cos Corr $\uparrow$
float16	1.86	1.00	1.00	1.00	1.00
fp8	3.47	1.00	0.95	0.95	1.00
int8	3.45	1.00	0.99	0.99	1.00
int4	6.29	1.00	0.91	0.91	1.00
int3	7.94	0.98	0.79	0.79	0.99
int2	11.41	0.91	0.54	0.53	0.95
binary	16.53	0.49	0.65	0.66	0.79

ID estimation on a 10,000-sample subset yields MLE $\hat{d} = 17.0$, TwoNN $\hat{d} = 12.6$, and LPCA $\hat{d} = 17.0$ (Figure 2a), confirming substantial structural redundancy relative to ambient dimension $d=1024$.

Table 1 summarizes quality across all methods. Int8 yields near-lossless retrieval (recall@10 = 0.99) at $3.5\times$ compression, matching the precision used by AlphaEarth and TESSERA [Brown et al., 2025, Feng et al., 2025]. Int4 provides a strong balanced operating point: $6.3\times$ compression with 91% kNN recall. Binary achieves the highest compression ($16.5\times$) with recall@10 of 0.65—suitable as a coarse first-pass filter before re-ranking with higher-fidelity representations [Wang et al., 2018, Jégou et al., 2011].

TurboQuant analysis. Table 2 and Figure 3 compare TurboQuant [Zandieh et al., 2025] against standard $\text{int}b$ at matched bit-widths. TurboQuant improves recall at every tested bit-width, but the gain shrinks rapidly as precision increases.

Largest gains at the lowest bit-widths. At 2 bits, turbo2 reaches recall@10 of 0.71 versus 0.54 for int2, a gain of +17% at essentially identical compression. At 3 bits, turbo3 improves recall from 0.79 to 0.88. The gap narrows at 4 bits (0.95 vs. 0.91) and is nearly zero at 8 bits (1.00 vs. 0.99).

Diminishing returns at higher precision. TurboQuant’s random orthogonal rotation redistributes variance across channels before scalar quantization. At 2–3 bits, this equalization materially improves range utilization and recall; once per-dimension affine quantization is already near-lossless (8 bits), the rotation has little room to help. The low intrinsic dimension of these embeddings ($\hat{d} \approx 13\text{--}17$) means 8-bit quantization already preserves neighborhood structure almost perfectly, confining rotation’s benefit to the most constrained operating points.

Practical recommendation. For geospatial embedding lakes, TurboQuant is most useful in the ultra-low-bit regime. If the deployment target is 2–3 bits, rotation offers a measurable quality gain; if 8-bit storage is acceptable, plain affine quantization is simpler and effectively equivalent.

Table 2: **TurboQuant [Zandieh et al., 2025] vs. standard *intb* on 49,785,116 embeddings.** Compression ratios are near-identical; TurboQuant improves recall most at low bit-widths. The advantage shrinks steadily as precision increases.

Method	Ratio ($\times$)	Recon. Cos $\uparrow$	C@10 $\uparrow$	E@10 $\uparrow$	Cos Corr $\uparrow$
int2	11.41	0.91	0.54	0.53	0.95
turbo2	11.44	0.91	0.71	0.69	0.99
int3	7.94	0.98	0.79	0.79	0.99
turbo3	7.96	0.98	0.88	0.88	1.00
int4	6.29	1.00	0.91	0.91	1.00
turbo4	6.10	1.00	0.95	0.95	1.00
int8	3.45	1.00	0.99	0.99	1.00
turbo8	3.45	1.00	1.00	1.00	1.00

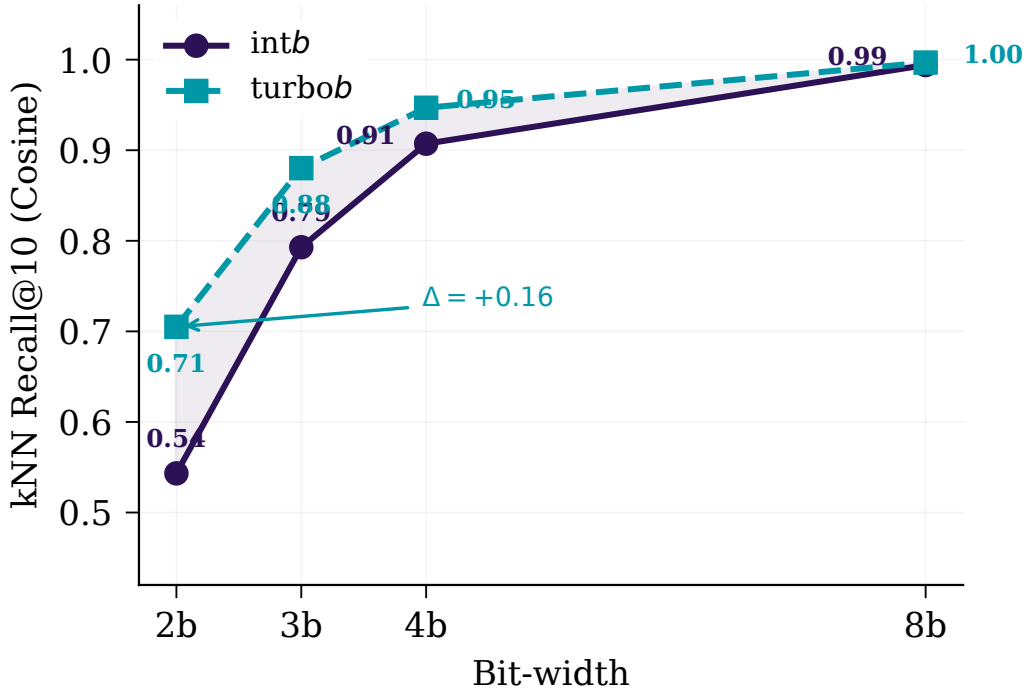


Figure 3: **Standard *intb* vs. TurboQuant recall@10 across bit-widths.** Cosine recall@10 across matched bit-widths. Shaded region marks the recall gap. TurboQuant improves recall at every matched bit-width, with the largest gain at 2 bits and a near-zero gap by 8 bits. Rotation helps most when quantization is extremely coarse, then becomes negligible once affine quantization is already near-lossless.

Improving binarization without retraining. To test whether binary quality can be improved in the strict post-hoc regime, we evaluate three lightweight variants: per-dimension median thresholding (binary_med), z-score thresholding (binary_zscore), and an ITQ-style PCA-whitened rotation (binary_itq). We run binary_med and binary_zscore on the full corpus; binary_itq is evaluated on the subsample only because its learned full-dimensional transform introduces large per-file metadata overhead in the current Parquet format. On a 20,000-embedding subsample, binary_med appears promising, improving cosine recall@10 from 0.637 to 0.659 at the same $32\times$ code-only compression. However, Figure 4 shows that this gain does *not* survive at corpus scale: on the full corpus, binary_med drops to 0.40 recall@10 and underperforms plain sign coding.

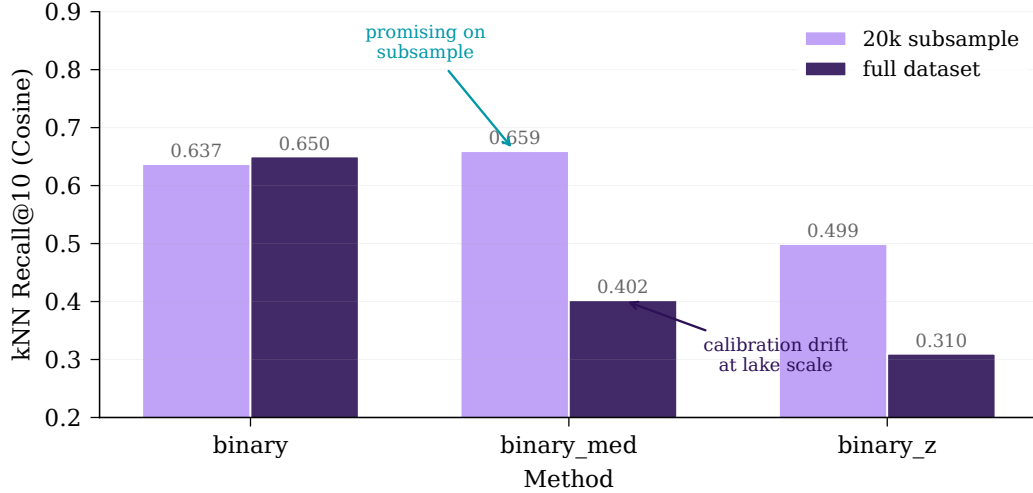


Figure 4: **Binary variants on subsample vs. full corpus.** `binary_med` improves over plain sign coding on a 20,000-embedding subsample, but the advantage reverses on the full corpus. This exposes a scale-sensitive failure mode: per-file threshold calibration does not transfer across partitions. Plain `binary` remains the strongest lake-scale binary baseline.

Both variants estimate thresholds independently within each Parquet partition, so the same semantic region of embedding space is binarized with different decision boundaries across geohash/year shards. That per-file calibration drift is mostly invisible on a local subsample but becomes destructive when nearest neighbors span many partitions. The result also reinforces a broader point from angular-hashing work [Gong et al., 2012]: preserving rank neighborhoods in binary space depends on consistent angular partitions, not merely on improving local reconstruction statistics. Indeed, `binary_zscore` attains very high reconstruction cosine (0.925) yet very poor recall@10 (0.310), showing that reconstruction fidelity is a poor proxy for binary retrieval quality.

Practically, the best-performing post-hoc binary method in our lake-scale setting is still the simplest one: global sign coding with no per-file calibration. This suggests that future binary improvements should use *global* calibration or shared transforms rather than independent per-partition thresholds.

Storage and throughput. Figure 1 shows the per-method storage footprint. Binary reduces the 183 GiB corpus to 11 GiB; `int4` to 29 GiB. The full pipeline (all 7 methods, 8 workers) processes the entire corpus in 45.2 min at 18,359 rows/s and 107 MiB/s output throughput.

On top of quantization, storage encoding yields another non-trivial reduction. Figure 5 summarizes the best Parquet-native and external backends on a shared 100,000-embedding sample. For `float32`, Zstd+BYTE_STREAM_SPLIT lowers storage from 30.17 to 27.44 effective bits per dimension, a further 9.0% reduction over plain Zstd. For `float16`, the same trick reduces storage from 15.41 to 14.28 bits/dim, a 7.4% gain. The effect does not transfer to low-bit packed codes: for `fp8`, `int4`, `int2`, and `binary`, BYTE_STREAM_SPLIT is neutral or harmful because the packed payload already mixes entropy across bit positions.

External codecs perform best when the payload is reorganized by dimension rather than by row. Dimension-major Zstd is the best external backend for `fp8`, `int4`, `int2`, and `binary`, reducing `binary` from 1.00 to 0.87 effective bits per original dimension on the sample. Pcodec [Loncaric et al., 2025] is strongest for higher-precision payloads: its dimension-major layout reaches 26.73 bits/dim for `float32`, 13.68 for `float16`, and 7.10 for `int8`, beating the best Parquet-native option in each case. Taken together, these results suggest a simple systems rule: use Parquet-native BYTE_STREAM_SPLIT only for floating-point embeddings, and prefer dimension-major external codecs if the storage stack can tolerate formats outside Parquet.

Discussion. First, ID estimates of 13–20 (vs. ambient $d=1024$) explain why aggressive quantization does not immediately collapse neighborhood structure: most per-dimension variance is redundant

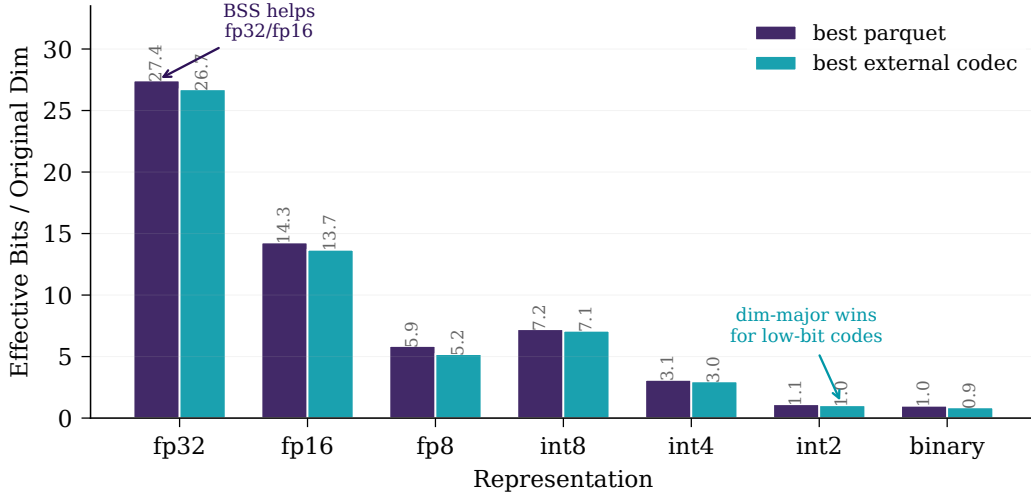


Figure 5: **Best Parquet-native vs. external storage backend on a 100,000-embedding sample.** Bars report effective bits per original dimension after lossless storage encoding. BYTE_STREAM_SPLIT helps only for high-precision floating-point payloads; for low-bit packed codes, dimension-major external compression is stronger.

for local geometry. Second, the quality decline from int8 to int2 is gradual (Figure 2b), enabling practitioners to select operating points based on their recall requirements versus storage budget. Third, binary quantization is not quality-preserving in a strict sense, but its $16.5\times$ compression and compatibility with Hamming-distance search make it attractive for first-pass candidate generation in multi-stage retrieval pipelines [Wang et al., 2018]. Fourth, post-hoc binary refinements that look promising on small subsamples can fail at lake scale (Figure 4); robustness across partitions matters more than local reconstruction quality, so future work should prefer shared/global calibration over per-file binary transforms. Fifth, storage layout remains a separate optimization lever after quantization: Parquet-native tricks help only for higher-precision floats, while low-bit packed representations benefit more from dimension-major external codecs than from additional Parquet encodings.

Retrieval benchmark. To validate that quantization-induced compression translates to practical search speedups, we benchmark brute-force kNN retrieval on a 1M-vector subset (1,000 queries, $k=10$) using FAISS [Johnson et al., 2021] on CPU and PyTorch cdist on a single NVIDIA RTX 3090 GPU. Each quantized corpus is dequantized to float32 before indexing, isolating the effect of quantization noise from format-specific index optimizations; binary embeddings use Hamming distance directly via packed uint8 search.

Table 3 and Figure 6 summarize throughput and recall. GPU search achieves 1,140–1,371 QPS for dequantized methods—a $22\times$ – $28\times$ speedup over FAISS CPU (48–60 QPS)—with matched recall, confirming that quantization quality is hardware-independent. Binary Hamming search runs at 270 QPS on CPU and 134 QPS on GPU; the lower GPU throughput reflects the overhead of bit-unpacking binary vectors into float tensors for cdist, a limitation of the brute-force approach that purpose-built Hamming kernels would eliminate.

Limitations. We use label-free neighborhood-preservation proxies rather than task-specific metrics (e.g., land-cover classification accuracy). kNN recall and cosine correlation are computed on a reservoir sample, not the full corpus. Our retrieval benchmark uses brute-force search; approximate nearest-neighbor indices such as inverted file (IVF) and hierarchical navigable small worlds (HNSW) may alter the quality–throughput frontier at larger scale [Johnson et al., 2021]. Future work should add labeled downstream evaluation (e.g., land-cover classification accuracy as in Chen et al. [2026]) and investigate ANN index interactions with low-bit quantization. Stacking post-hoc quantization on top of learned discrete representations [Mentzer et al., 2024, Chen et al., 2026] may compound compression gains further.

Table 3: **Brute-force search benchmark** on 1M embeddings (1K queries, $k=10$). QPS = queries per second. GPU is a single RTX 3090. Recall is identical across backends since quantization quality is hardware-independent. **Bold**: best throughput per row.

Method	CPU QPS	GPU QPS	Speedup	R@10
float16	60	1322	22.2×	1.00
fp8	49	1238	25.5×	0.95
int8	49	1140	23.5×	0.99
int4	48	1371	28.3×	0.90
int3	48	1224	25.3×	0.81
int2	48	1157	24.4×	0.61
binary	270	134	0.5×	1.00*

*Recall measured against binary GT (Hamming distance).

5 Conclusion

TerraBit provides a systematic compression study for geospatial embedding lakes. On 49.8M real embeddings, low-bit quantization yields a smooth quality–cost frontier: int4/int8 are strong balanced choices; binary is viable for coarse retrieval stages. GPU-accelerated brute-force search over dequantized embeddings exceeds 1,300 QPS on a single RTX 3090, demonstrating that quantized embedding lakes are practical for real-time retrieval at scale.

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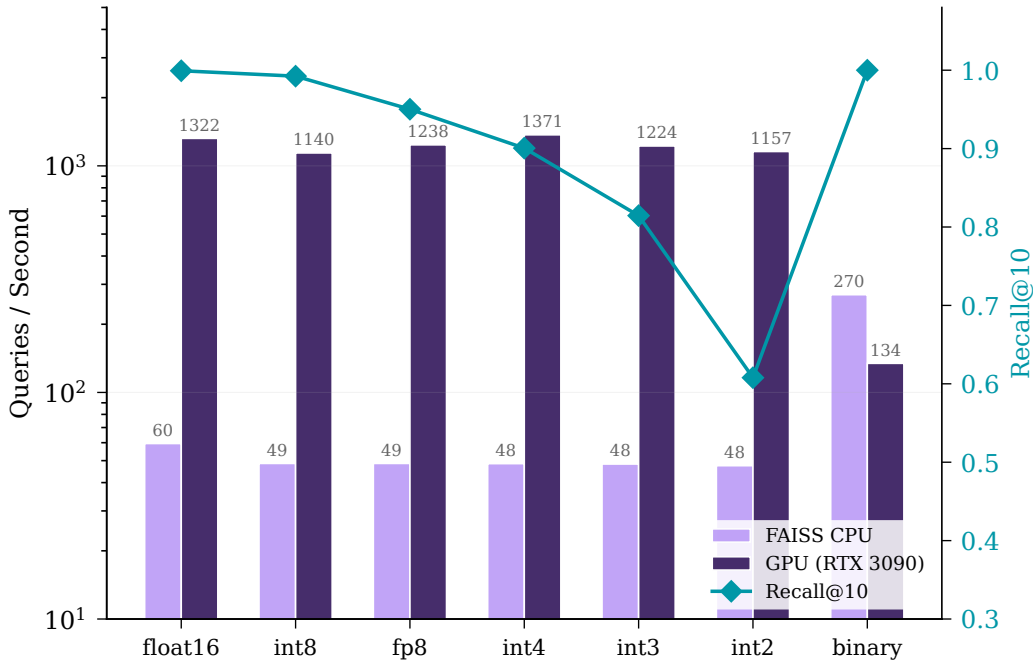


Figure 6: **Search throughput vs. recall** on 1M embeddings. GPU brute-force delivers 1,140–1,371 QPS across dequantized methods. Binary Hamming search is fastest on CPU (270 QPS) due to native bitwise ops. Recall (purple diamonds, right axis) tracks quantization fidelity independently of hardware.

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